

AWS Lab Setup #4

1. System Design

The EVIL Bank system was designed by AWS to meet the requirements for high availability, scalability, security and disaster recovery.

A Virtual Private Cloud was created with a CIDR block 10.0.0.0/16.

An Internet Gateway was attached to allow public access and NAT Gateway was used to allow private instances to access the internet securely.

2. Scalability

It is achieved by using an Application Load Balancer, which is used to distribute incoming traffic across the EC2 instances allowing the system to handle an increase in traffic.

3. High Availability

High availability is ensured by using a Load balancer to distribute the traffic, using multiple Availability zones and deploying subnets

4. Security

Security Groups control traffic, private subnets prevent direct access to EC2 instances on the back end and direct access to EC2 is restricted.

5. Remote access

Remote access to the instances was performed using SSH (EC2 Instance Connect) allowing the administrators to manage the system in a secure matter.

6. Disaster recovery

It is supported through scalable infrastructure, recreate instances quickly and Multi-AZ deployment and this helps with ensuring minimal downtime in the chances of failures.

7. Implementation Summary

These AWS services below were configured

- 2 public and 2 private subnets
- Route tables
- Internet and NAT gateway

- Security groups
- Target group
- Running Apache web server
- Application Load Balancer

8. Testing and Validation

- Health check validation cleared
- The application was accessed through Load Balancer DNS and displayed EVIL Bank Running on the console
- Only the Load Balancer is able to be publicly accessed and the EC2 instance is not directly accessible
- The Load Balancer also successfully routed traffic to the EC2 instance without errors.

9. Conclusion

The EVIL Bank architecture meets all required objectives, including security, disaster recovery, high availability and scalability. The system is accessible through the Load balancer.

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subnets | VPC | us-east-2

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AWS Lab Setup #4

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Loza Kidanemariam (6348-8247-3376)

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VPC > Subnets

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

Subnets (7) Info

Last updated 5 minutes ago

Actions

Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IPv4 C
☐	-	subnet-0c36cf26a2f6044ce	Available	vpc-0bdef5a19da6a709b	Off	172.31
☐	-	subnet-0cc053584e6c19a27	Available	vpc-0bdef5a19da6a709b	Off	172.31
☐	-	subnet-0383b799e2fdb5e8b	Available	vpc-0bdef5a19da6a709b	Off	172.31
☐	Public-1	subnet-0126a80e9fc9fb771	Available	vpc-0df78a09cdefd65a1 EvilB...	Off	10.0.1
☐	Public-2	subnet-0a82e93d13c977714	Available	vpc-0df78a09cdefd65a1 EvilB...	Off	10.0.2
☐	Private-1	subnet-09195b4a7c195b0b1	Available	vpc-0df78a09cdefd65a1 EvilB...	Off	10.0.3
☐	Private-2	subnet-032e7a0320a3247cd	Available	vpc-0df78a09cdefd65a1 EvilB...	Off	10.0.4

Select a subnet

subnets

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VPC | us-east-2

Internet gateways > igw-01053b2ab89185a86

VPC dashboard

Filter by VPC

- Virtual private cloud
 - Your VPCs
 - Subnets
 - Route tables
 - Internet gateways
 - Egress-only internet gateways
 - DHCP option sets
 - Elastic IPs
 - Managed prefix lists
 - NAT gateways
 - Peering connections
 - Route servers
- Security
 - Network ACLs
 - Security groups
- PrivateLink and Lattice
 - Getting started
 - Endpoints
 - Endpoint services

igw-01053b2ab89185a86 / EvilBank-IGW

Details info

Internet gateway ID: igw-01053b2ab89185a86

State: Attached

VPC ID: vpc-0df78a09cdefd65a1 | EvilBank-VPC

Owner: 634882473376

Tags (1)

Key	Value
Name	EvilBank-IGW

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VPC | us-east-2

Route tables > rtb-011b7c5a99eca43f6

VPC dashboard

Filter by VPC

- Virtual private cloud
 - Your VPCs
 - Subnets
 - Route tables
 - Internet gateways
 - Egress-only internet gateways
 - DHCP option sets
 - Elastic IPs
 - Managed prefix lists
 - NAT gateways
 - Peering connections
 - Route servers
- Security
 - Network ACLs
 - Security groups
- PrivateLink and Lattice
 - Getting started
 - Endpoints
 - Endpoint services

rtb-011b7c5a99eca43f6 / Public-RT

Updated routes for rtb-011b7c5a99eca43f6 / Public-RT successfully

Details

Route table ID: rtb-011b7c5a99eca43f6

Main: No

Explicit subnet associations: -

Edge associations: -

VPC: vpc-0df78a09cdefd65a1 | EvilBank-VPC

Owner ID: 634882473376

Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-01053b2ab89185a86	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

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VPC

NAT gateways

nat-0d86465f220dc1a39

VPC dashboard

Virtual private cloud

Security

PrivateLink and Lattice

Details

NAT gateway ID

nat-0d86465f220dc1a39

NAT gateway ARN

arn:aws:ec2:us-east-2:634882473376:natgateway/nat-0d86465f220dc1a39

VPC

vpc-0df78a09cdefd65a1 / EvilBank-VPC

Connectivity type

Public

Primary public IPv4 address

3.138.90.142

Subnet

subnet-0126a80e9fc9fb771 / Public-1

State

Available

Primary private IPv4 address

10.0.1.232

Created

Monday, April 13, 2026 at 17:29:40 P DT

State message

-

Primary network interface ID

eni-0ed52a7b97928fe07

Deleted

-

Secondary IPv4 addresses

Monitoring

Attachments

Tags

Secondary IPv4 addresses

Private IPv4 address

Network interface ID

Status

Failure message

Secondary IPv4 addresses are not available for this nat gateway.

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VPC

Route tables

rtb-01e80dbbfb8153952

VPC dashboard

Virtual private cloud

Security

PrivateLink and Lattice

Updated routes for rtb-01e80dbbfb8153952 / Private-RT successfully

Details

rtb-01e80dbbfb8153952 / Private-RT

Actions

Details

Route table ID

rtb-01e80dbbfb8153952

VPC

vpc-0df78a09cdefd65a1 | EvilBank-VPC

Main

No

Explicit subnet associations

-

Edge associations

-

Owner ID

634882473376

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Both

Edit routes

Destination

Target

Status

Propagated

Route Origin

0.0.0.0/0

nat-0d86465f220dc1a39

Active

No

Create Route

10.0.0.0/16

local

Active

No

Create Route Table

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SecurityGroup | EC2 | us-east-2

SecurityGroup | EC2 | us-east-2

sg-02bb996fa9d742929 - ALB-SG

Security group (sg-02bb996fa9d742929 | ALB-SG) was created successfully

Details

sg-02bb996fa9d742929 - ALB-SG

Details

Security group name: ALB-SG

Security group ID: sg-02bb996fa9d742929

Description: Load Balancer Security Group

VPC ID: vpc-0df78a09cdefd65a1

Owner: 634882473376

Inbound rules count: 1 Permission entry

Outbound rules count: 1 Permission entry

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0c89160e72c66d83f	IPv4	HTTP	TCP	80

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CreateSecurityGroup | EC2 | us-east-2

Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name: EC2-SG

Description: EC2 Web Server Security Group

VPC: vpc-0df78a09cdefd65a1 (EvilBank-VPC)

Inbound rules

Type	Protocol	Port range	Source	Description - optional
HTTP	TCP	80	Custom sg-02bb996fa9d742929	
SSH	TCP	22	My IP 172.56.186.180/32	

Outbound rules

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Instances | EC2 | us-east-2

EC2 Instance Connect | us... AWS Lab Setup Help EC2 Instance Connect | us... Untitled document - Goog... AWS Lab Setup #4

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EC2 > Instances

Dashboard AWS Global View Events

▼ Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager

▼ Images

- AMIs
- AMI Catalog

▼ Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

▼ Network & Security

- Security Groups

Instances (1/4) Info Last updated 10 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Pub
☐	QuickBite-Server	i-0c1d48e1fb67d7fa3	Running	t3.micro	3/3 checks passed	View alarms	us-east-2c	ec2-
☐	NoteHub-Serv...	i-03e52e25a91491550	Running	t3.micro	3/3 checks passed	View alarms	us-east-2c	ec2-
☐	NoteHub-Serv...	i-0672f80c458e4023e	Running	t3.micro	3/3 checks passed	View alarms	us-east-2c	ec2-
☒	Temp-Server	i-051b65e23e85cbd13	Running	t3.micro	3/3 checks passed	View alarms	us-east-2a	-

i-051b65e23e85cbd13 (Temp-Server)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID i-051b65e23e85cbd13

Public IPv4 address 18.224.64.163 | open address

Private IPv4 addresses 10.0.1.229

IPv6 address -

Instance state Running

Public DNS -

Hostname type IP name: ip-10-0-1-229.us-east-2.compute.internal

Private IP DNS name (IPv4 only) ip-10-0-1-229.us-east-2.compute.internal

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Instance details | EC2 | us... EC2 Instance Connect | u... AWS Lab Setup Help EC2 Instance Connect | us... Untitled document - Goog... AWS Lab Setup #4

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Available Versions:

Version 2023.11.20260413:

Run the following command to upgrade to 2023.11.20260413:

```
dnf upgrade --releasever=2023.11.20260413
```

Release notes:

<https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.11.20260413.html>

Installed:

```
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.2.x86_64      apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch      httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64
```

Complete!

```
[ec2-user@ip-10-0-1-229 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:00:23 ago on Tue Apr 14 04:34:33 2026.
Package httpd-2.4.66-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-1-229 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-1-229 ~]$ echo "EVIL Bank Running" | sudo tee /var/www/html/index.html
EVIL Bank Running
[ec2-user@ip-10-0-1-229 ~]$ curl localhost
curl: (7) Failed to connect to localhost port 80 after 0 ms: Could not connect to server
[ec2-user@ip-10-0-1-229 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-1-229 ~]$ curl localhost
EVIL Bank Running
[ec2-user@ip-10-0-1-229 ~]$
```

i-051b65e23e85cbd13 (Temp-Server)

PublicIPs: 18.224.64.163 PrivateIPs: 10.0.1.229

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Load balancer details | EC2...AWS Lab Setup HelpEC2 Instance Connect | us...EC2 Instance Connect | us...Untitled document - Goog...AWS Lab Setup #4

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EC2 > Load balancers > EvilBank-ALB

DashboardAWS Global View iEvents

InstancesInstancesInstance TypesLaunch TemplatesSpot RequestsSavings PlansReserved InstancesDedicated HostsCapacity ReservationsCapacity Manager New

ImagesAMIsAMI Catalog

Elastic Block StoreVolumesSnapshotsLifecycle Manager

Network & SecuritySecurity Groups

EvilBank-ALB

Details

Load balancer typeApplication

SchemeInternet-facing

Load balancer ARNarn:aws:elasticloadbalancing:us-east-2:634882473376:loadbalancer/app/EvilBank-ALB/c40852dbd6cc0f67

StatusActive

Hosted zoneZ3AADJGX6KCTL2

DNS name infoEvilBank-ALB-2011027736.us-east-2.elb.amazonaws.com (A Record)

VPCvpc-0df78a09cdefd65a1

Availability Zonessubnet-0126a80e9fc9fb771 us-east-2a (use2-az1)subnet-0a82e93d13c977714 us-east-2b (use2-az2)

Load balancer IP address typeIPv4

Date createdApril 13, 2026, 22:10 (UTC-07:00)

Listeners and rules

Network mappingResource mapSecurityMonitoringIntegrationsAttributesCapacityTags

Listeners and rules (1) info

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

Protocol:PortDefault actionRulesLoad balancer detailsARNSecurity policyDefault SSL

HTTP:80

Forward to target groupEvilBank-TG 1 (100%)Target group stickiness: Off1 ruleARNNot applicableNot applicab

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Target group details | ...AWS Lab Setup HelpEC2 Instance Conne...EC2 Instance Conne...Untitled document - ...AWS Lab Setup #4evilbank-alb-201102...

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EC2 > Target groups > EvilBank-TG

Capacity Manager New

ImagesAMIsAMI Catalog

Elastic Block StoreVolumesSnapshotsLifecycle Manager

Network & SecuritySecurity GroupsElastic IPsPlacement GroupsKey PairsNetwork Interfaces

Load BalancingLoad BalancersTarget GroupsTrust Stores

Auto ScalingAuto Scaling Groups

Settings

EvilBank-TG

Details

arn:aws:elasticloadbalancing:us-east-2:634882473376:targetgroup/EvilBank-TG/827ad5f4518bc51b

Target typeInstance

IP address typeIPv4

Protocol : PortHTTP: 80

Load balancerEvilBank-ALB

Protocol versionHTTP1

VPCvpc-0df78a09cdefd65a1

1Total targets

1Healthy

0Unhealthy

0Unused

0Initial

0Draining

0Anomalous

Distribution of targets by availability zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

MonitoringHealth checksAttributesTags

Registered targets (1) info

Anomaly mitigation: Not applicableDeregisterRegister targets

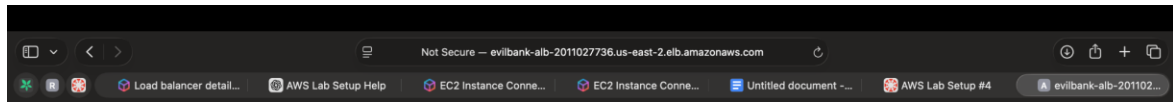
Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

Instance IDNamePortZoneHealth statusHealth status detailsAdmini...

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EVIL Bank Running